HYPOTHESIS TESTING SUMMARY

ApexPlanet Data Analytics Internship — Task 4

1. Business Question

Do average transaction sales differ across the product categories in the supplied sales dataset? This test checks whether the observed category-level differences in transaction sales are statistically significant.

2. Hypotheses

Null hypothesis (H■): The population mean transaction sales are equal across all product categories.
Alternative hypothesis (H■): At least one product category has a different population mean transaction sales value.

3. Statistical Method

Test used: One-way Analysis of Variance (ANOVA).
Significance level (α): 0.05 (5%).
Why this test: One-way ANOVA compares the means of a numeric outcome across three or more independent groups. Here, the numeric outcome is Total_Sales and the groups are product categories.

4. Results

Measure	Result
Number of records with valid sales	1,000
Number of categories tested	5
F-statistic	0.7171
p-value	0.580286
Significance level	0.05
Decision	Fail to reject the null hypothesis

5. Category-Level Descriptive Statistics

The table below provides the sample size and average transaction sales for each category. These descriptive statistics help explain the data, but they do not replace the hypothesis test.

Category	Records (n)	Mean sales	Std. deviation
Grocery	153	145,305.30	114,918.09
Electronics	354	143,442.32	117,601.79
Education	178	140,627.47	116,852.56
Furniture	159	135,355.73	111,048.77
Fashion	156	127,153.18	105,149.54

6. Interpretation and Business Conclusion

The test does not provide sufficient evidence at the 5% significance level that mean transaction sales differ across categories.

A p-value is the probability, assuming the null hypothesis and model assumptions are true, of obtaining an F-statistic at least as extreme as the observed one. If $p < 0.05$, reject H■; if $p \geq 0.05$, fail to reject H■. Failing to reject H■ does not prove that all category means are identical.

7. Limitations and Next Steps

• ANOVA assumes independent observations and is sensitive to unequal variances and strongly non-normal group distributions.
• This is an observational comparison, not an A/B experiment; it does not establish that category causes higher or lower sales.
• Check group variances and residuals; if assumptions are not suitable, consider Welch’s ANOVA or a non-parametric alternative.
• For business decisions, also compare order volume, units sold, profit margin, and confidence intervals. If ANOVA is significant, use an appropriate post-hoc test to identify which categories differ.

Source: user-provided ApexPlanet sales dataset. Prepared for Task 4.